Math 152 Exam III questions 4/29

Use series: $\lim_{x \to 0} \dfrac{x^2 - \tan^{-1}(x^2)}{e^{x^3} - 1 - x^3}$

$$e^{x^3} = \sum_{n=0}^{\infty} \frac{(x^3)^n}{n!} = 1 + x^3 + \frac{x^6}{2!} + \frac{x^9}{3!} + \frac{x^{12}}{4!} \cdots$$

$$\tan^{-1}(x^2) = \sum_{n=0}^{\infty} \frac{(-1)^n (x^2)^{2n+1}}{2n+1} \quad \frac{x^2}{1} - \frac{x^6}{3} + \frac{x^{10}}{5} - \frac{x^{14}}{7}$$

$$\lim_{x \to 0} \frac{x^2 - \left(x^2 - \frac{x^6}{3} + \frac{x^{10}}{5} - \frac{x^{14}}{7} + \cdots\right)}{1 + x^3 + \frac{x^6}{2!} + \frac{x^9}{3!} + \frac{x^{12}}{4!} \cdots - 1 - x^3}$$

$$\lim_{x \to 0} \left(\frac{\frac{x^6}{3} - \frac{x^{10}}{5} + \frac{x^{14}}{7} \cdots}{\frac{x^6}{2!} + \frac{x^9}{3!} + \frac{x^{12}}{4!} \cdots} \right)$$

$$\lim_{x \to 0} \frac{\cancel{x^6}\left(\frac{1}{3} - \frac{x^4}{5} + \frac{x^8}{7} \cdots\right)}{\cancel{x^6}\left(\frac{1}{2!} + \frac{x^3}{3!} + \frac{x^6}{4!} \cdots\right)} = \frac{\frac{1}{3} + 0 + 0}{\frac{1}{2} + 0 + 0 + 0}$$

$$= \frac{2}{3}$$

Find R $\displaystyle\sum_{n=0}^{\infty} \frac{(-1)^n x^n}{4^n n!}$

Ratio Test

$$\lim_{n\to\infty} \left| \frac{x \cdot \cancel{x^n}}{4 \cdot \cancel{4^n} (n+1) \cancel{n!}} \cdot \frac{\cancel{4^n}\,\cancel{n!}}{\cancel{x^n}} \right|$$

$$a_n = \frac{x^n}{4^n n!}$$

$$a_{n+1} = \frac{x^{n+1}}{4^{n+1}(n+1)!}$$

$$\frac{x \cdot x^n}{4 \cdot 4^n (n+1) n!}$$

$$\lim_{n\to\infty} \left| \frac{x}{4(n+1)} \right| = 0 < 1$$

always

$R = \infty$

Find the $T_3(x)$ $f(x) = \sin(x)$ at $\pi/6$

$f(x) = \sin x$ $\quad f(\pi/6) = \frac{1}{2}$

$f'(x) = \cos x$ $\quad f'(\pi/6) = \sqrt{3}/2$

$f''(x) = -\sin x$ $\quad f''(\pi/6) = -1/2$

$f'''(x) = -\cos x$ $\quad f'''(\pi/6) = -\sqrt{3}/2$

$f''''(x) = \sin x$ $\quad \cancel{f''''(\pi/6)}$

$$\sin(\pi/6) \approx \frac{1}{2} + \frac{\sqrt{3}}{2}(x - \pi/6) - \frac{1}{2}\frac{(x-\pi/6)^2}{2!} - \frac{\sqrt{3}}{2}\frac{(x-\pi/6)^3}{3!}$$

$$T_3(x) = S_4$$

Find the directrix for $x^2-4x-4y=0$

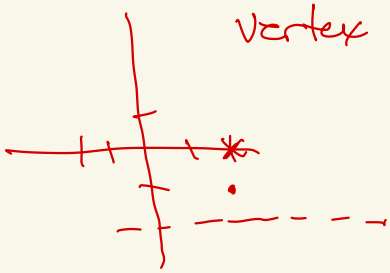
$y=-2$

$x^2-4x+4=4y+4$

$(x-2)^2=4(y+1)$

Vertex $(2,-1)$ $4p=4$

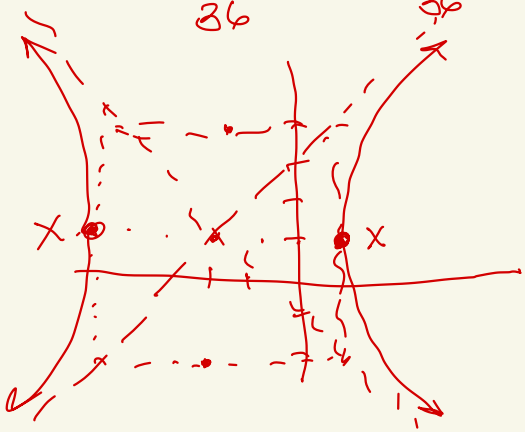
$p=1$



$$\frac{4(x+2)^2}{36}-\frac{4(y-1)^2}{36}=\frac{36}{36}$$

$$\boxed{\frac{(x+2)^2}{9}-\frac{(y-1)^2}{9}=1}$$

$a=3$ $b=3$



center $(-2,1)$

vertices $(-2\pm3,1)$

$c^2=a^2+b^2$

$c^2=9+9=18$ $3\sqrt{2}=c$

foci $=(-2\pm3\sqrt{2},1)$

$(y-1)=\pm(x+2)$

Binomial Series

$$(1+\bigcirc)^k \qquad \sum_{n=0}^{\infty} \frac{\square\,(\bigcirc)^n}{n!}$$

$$\square = 1 + k + k(k-1) + k(k-1)(k-2) + k(k-1)(k-2)(k-3)$$

$$f(x) = (x+1)^{1/2} = (1+x)^{1/2} = \sum_{n=0}^{\infty} \square \frac{x^n}{n!}$$

$$\square = 1 + \frac{1}{2} + \frac{1}{2}\left(\frac{-1}{2}\right) + \frac{1}{2}\left(\frac{-1}{2}\right)\left(\frac{-3}{2}\right) + \frac{1}{2}\left(\frac{-1}{2}\right)\left(\frac{-3}{2}\right)\left(\frac{-5}{2}\right) \ldots$$

$$\underset{n=0}{1} + \underset{n=1}{\frac{1}{2}} - \underset{n=2}{\frac{1}{2^2}} + \underset{n=3}{\frac{1\cdot 3}{2^3}} - \underset{n=4}{\frac{1\cdot 3\cdot 5}{2^4}} + \underset{n=5}{\frac{1\cdot 3\cdot 5\cdot 7}{2^5}} \ldots$$

$$\frac{(-1)^{n+1}(1\cdot 3\cdot 5\cdot 7 \cdots (2n-3))}{2^n}$$

can't be neg

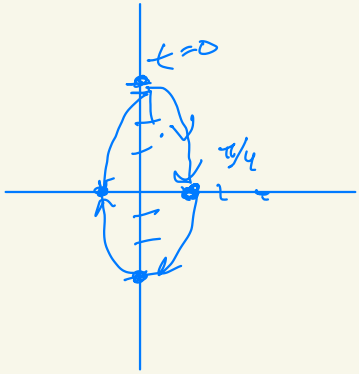
$$1 + \frac{1}{2}\frac{x}{1!} + \sum_{n=2}^{\infty} \frac{(-1)^{n+1}(1\cdot 3\cdot 5 \cdots (2n-3))}{2^n\, n!} x^n$$

Parametric

$x = \sin 2\theta \quad y = 3\cos 2\theta$

t	x	y
0	0	3
$\pi/4$	1	0

clockwise



$$f(x) = \frac{1}{4+x} = \frac{1/4}{1-\left(-\frac{x}{4}\right)}$$